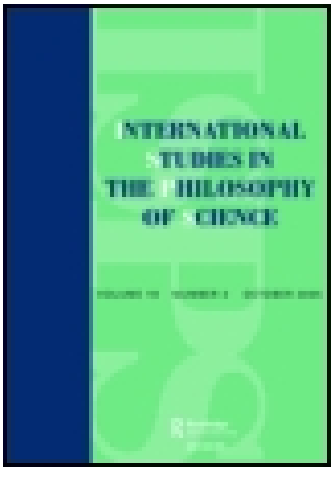
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Reassessing the Prospects for a Growing Block Model of the Universe

John Earman

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Although C. D. Broad's notion of Becoming has received a fair amount of attention in the philosophy-of-time literature, there are no serious attempts to show how to replace the standard 'block' spacetime models by models that are more congenial to Broad's idea that the sum total of existence is continuously increased by Becoming or the coming into existence of events. In the Newtonian setting Broad-type models can be constructed in a cheating fashion by starting with a Newtonian block model, carving chips off the block, and assembling the chips in an appropriately structured way. However, attempts to construct Broad-type models in a non-cheating fashion reveal a number of problematic aspects of Becoming that have not received adequate attention in the literature. The paper then turns to an assessment of the problem and prospects of adapting Becoming models to relativistic spacetimes. The results of the assessment differ in both minor and major ways from the ones in the extant literature. Finally, the paper describes how the causal set approach to quantum gravity promises to provide a mechanism for realizing Becoming, though the form of Becoming that emerges may not conform to any of the versions discussed in the philosophical literature.

1. Introduction

The philosophy of time is filled with discontents. One perennial discontent focuses on the spacetime models provided by Newtonian, special relativistic, and general relativistic physics alike. An often voiced complaint is that these models all fail to capture an important dynamical aspect of time by presenting a purely static (a.k.a. 'block universe') representation of events.[1] The philosophical literature contains many purported remedies for this perceived defect, but the two main lines of inquiry—associated, respectively, with J. M. E. McTaggart and C. D. Broad—can be briefly (if somewhat crudely) characterized as follows. The first line tries to animate the standard

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block models by supplementing the *B*-series—the ordering of events with respect to earlier and later—with an *A*-series, the ordering of events with respect to the intrinsic but transitory properties of being past, being present, and being future.[2] McTaggart (1908) is (in)famous for arguing that (1) time requires change; (2) the *A*-series is essential to real change; and (3) the *A*-series is conceptually incoherent, and, thus, time and change are ideal. Later writers have taken issue with McTaggart's argument for (3), and some of them are hopeful that the first route to a dynamical conception of time remains open.[3] The second route, pioneered by C. D. Broad (1923), does not seek to achieve dynamism by adding *A*-series properties or other animating properties to block models but rather by replacing the standard block models by growing block models. A related but different route to a dynamical theory of time involving branching spacetime structures and branch attrition is explored by Storrs McCall (1994). McCall's proposal will not be discussed here, but some skeptical remarks about branching spacetimes can be found in Earman (2008).

Broad's approach languished for several decades in a philosophical backwater. But it was revived and vigorously defended in Michael Tooley's book *Time, Tense, and Causation* (1997). Such a revival is perhaps not surprising, since philosophical ideas have a tendency to recycle in popularity. But it is interesting that recently some physicists have begun to espouse versions of the growing block model (see Ellis 2006; Sorkin 2007). The purpose of this paper is to assess the viability of the growing block model. I will not pronounce on the relative merits of static vs. dynamical accounts of time or on property-based vs. non-property-based versions of a dynamical account of time.[4] Admittedly, a full assessment of the growing block model could only be reached in the context of these more general issues, a full discussion of which would require a book-length treatment. But the main points I wish to make can be stated and evaluated without having to settle these general (and perhaps intractable) issues.

The plan of the paper is as follows. Section 2 provides a brief review of Broad's conception of Becoming. Section 3 clarifies what is meant by a block model for present purposes. Section 4 shows how Broad's conception of Becoming can be implemented in the Newtonian setting in such a way that the growing block model avoids some of the embarrassments that are often claimed to arise. Admittedly, the proposed growing block model does not provide the kind of animation desired by some proponents of a dynamical theory of time, but I take this not as a criticism of the model but simply a way of underscoring the point that Becoming and animation need to be distinguished. My worry about the proposed growing block model is that if it is to avoid parasitism on the block model while exhibiting desired features, it must rely on a number of substantive assumptions such as determinism and the prohibition of stationary or strictly periodic *B*-series scenarios. Section 5 discusses additional problems that beset Broad's version of the growing block model in a general relativistic setting. Section 6 reviews the difficulties that a different, non-Broadian form of the growing block model encounters in the same setting. While Broad's views on the nature of existence are not subject to outright refutation, the discussion to this juncture indicates that there is a considerable price to be paid for maintaining them. But, as discussed in Section 7, one of the current research programmes aimed at producing a quantum theory of gravity—the causal

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set approach—revives the fortunes of the growing block model, albeit in a way that may not conform to Broad's vision of how the block grows. While the fate of this research programme—or any other scientific research programme for that matter—cannot settle metaphysical debates about the nature of time and existence, it can inform and help to shape the debates, and, consequently, metaphysicians would be foolish to ignore the ideas offered by these programmes. Conclusions are contained in Section 8.

2. Broad and the Growing Block Universe

The growing block model was first set out in Broad's *Scientific Thought* (1923). Although he continued to use much the same terminology in the later work, *Examination of McTaggart's Philosophy* (1938), his views on the nature of time and change shifted significantly (see also Broad 1959).[5] I will not attempt to chart this shift and will be concerned here only with the views in *Scientific Thought*.

Here is a characteristic passage from *Scientific Thought* setting forth Broad's contention that the growth of the growing block model grounds the *B*-series and also gives it a directionality.[6]

> The sum total of existence is always increasing, and it is this which gives the time-series a sense as well as an order. A moment t is later than a moment t' of the sum total of existence at t includes the sum total of existence at t' together with something more. (Broad 1923, 66–67)

For Broad, a change from future to present is not a matter of an event losing the *A*-series property of futurity and taking on the property of presentness; rather, it is a matter of the event 'becoming' or coming into existence.

> [W]hen an event becomes, *it comes into existence*; and it was not anything at all until it had become … The relation between existence and becoming is very intimate. Whatever is has become, and the sum total of the existent is continually augmented by becoming. There is no such thing as *ceasing* to exist; what has become exists henceforth for ever. When we say that something has ceased to exist we only mean that it has ceased to be present; and this only means that the sum total of existence has increased since any part of the history of the thing became. (Broad 1923, 68–69)

McTaggart lived long enough to have read Broad's *Scientific Thought* (1923), but as far as I am aware he never commented on Broad's growing block model.[7] But in 'The Unreality of Time'(1908) he flirted with a non-property-based dynamical time, only to shy away from it for a not very good reason.

> Could we say that, in a time which formed a *B*-series but not an *A*-series, the change consisted in the fact that an event ceased to be an event, while another event began to be an event? If this were the case, we should certainly have got change. But this is impossible. An event can never cease to be an event. It can never get out of any series in which it once is. (McTaggart 1908, 459)

Broad could not have agreed more that events can never cease to be events and, consequently, that Presentism, which entails the unreality of the past as well as the future, is not a viable option. Where he parted company with McTaggart is with McTaggart's leap to the conclusion that

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> Since, therefore, what occurs in time never begins or ceases to be, or to be in itself, and since, again, if there is change it must be change of what occurs in time … only one alternative remains. Change must happen to the events of such a nature that the occurrence of these changes does not hinder these events from being events, and the very same events, both before and after the change. (McTaggart 1908, 460)

But (Broad might have asked) why is the only alternative to *A*-series change the unacceptable idea of event annihilation? Why must there be a symmetry between annihilation and creation? Why can't events Become or come into existence but never be annihilated or go out of existence? It is hard to resist the conjecture that reading this passage in 'The Unreality of Time' would have naturally led Broad to ask these questions and answer them with his growing block model.

Despite some attractive features, Broad's growing block model is subject to a number of qualms. One derives from the fact that the model seems to be based on the idea that the layers of existence successively coming into being are slices of Newtonian absolute time, leading one to wonder how the model can be reconciled with the developments of the special and general theories of relativity. There are also attempts to embarrass the growing block model, even in the Newtonian setting, with such rhetorical questions as these: How fast is the block growing? With respect to what is the block growing—some additional dimension? Is this additional dimension an extra dimension of spacetime or a dimension external to spacetime? If the former, how can it be incorporated into the spacetime models of physics without compromising empirical adequacy? If the latter, what role can it play in the explanation of events in the spacetime nexus? Before these and other questions about the growing block model can be tackled, we need a more detailed specification of how the growing block model is constituted and how it functions to support the sort of semantics and metaphysics Broad thought flowed from Becoming. Typically what we get in the literature is a diagram showing a sequence of inked columns of increasing height, which is supposed to illustrate the growth of the growing block. A picture may be worth a thousand words, but in this case those words don't speak clearly enough to provide a basis for assessing the viability of Broad's model.

In what follows I will refer to those who advocate standard block models as providing adequate representations of time as Blockheads (no insult intended)[8] and to those who advocate Broad's growing block model, or some variant thereof, as Broadheads.

3. Block Models

All of the block spacetime models—whether Newtonian, special relativistic, or general relativistic—have a common form, *viz.* $\langle M, G_1, G_2, \ldots, P_1, P_2, \ldots\rangle$, where M is a differentiable manifold; and the G_i and P_j are geometric object fields (scalar fields, vector fields, tensor fields, etc.) on M, the intended interpretation being that the G_i characterize the geometric structure of the spacetime, while the P_j characterize the matter–energy contents of spacetime. In both the Newtonian and special relativistic cases it is assumed that there is a unique (up to isomorphism)[9] spacetime background common to all physically possible models. In the case of the special theory of relativity

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(STR) that background is Minkowski spacetime, where $M = \mathbb{R}^4$ and $G_1 = \eta_{ab}$ (the Minkowski metric), all the other G_i (e.g. the affine connection) being definable in terms of η_{ab}. In the case of Newtonian theory M is also assumed to be $\mathbb{R}^4$, but a larger number of geometric object fields are needed to characterize the structure of this spacetime; in particular, there is a distinguished global time function, a space metric, a time metric, and an affine connection, and there are various mathematical conditions that ensure proper meshing of these objects. The details are not important here.[10] It suffices for present purposes to focus on the distinguished global time function, which is a smooth map $T : M \to \mathbb{R}$, whose level surfaces $T = const$ define the planes of absolute Newtonian simultaneity. The range of T is $(-\infty, +\infty)$, with larger values indicating later times. It is also assumed here that T is calibrated with the time metric, in that the value of $|T(p_1) - T(p_2)|$ coincides with the measure of the temporal separation of $p_1, p_2 \in M$.

In the case of the general theory of relativity (GTR) the spacetime structure is not fixed but varies from model to model, and in contrast to the STR the spacetime manifold M may have a nontrivial topology and the spacetime metric g_{ab} need not be pseudo-Euclidean. The basic postulate of GTR is that the spacetime metric g_{ab} and the stress-energy tensor T_{ab} (which is a functional of the matter fields described by the P_j) are jointly constrained by Einstein's field equations, a set of nonlinear partial differential equations, the details of which are irrelevant here.

It will be assumed that any admissible block model $\langle M, G_1, G_2, \ldots, P_1, P_2, \ldots\rangle$ is inextendable or, equivalently, that it is not the truncation of some other block model $\langle M', G'_1, G'_2, \ldots, P'_1, P'_2, \ldots\rangle$, i.e. there is no diffeomorphism $d : M \to M'$ such that $d(M)$ is a proper subset of M' and $d^*G_i = G'_i$ and $d^*P_j = P'_j$ for all i and j. Advocates of Becoming would presumably not object to this condition; indeed, they would presumably endorse their own version of it in the guise of the postulate that Becoming does not come to a halt unless there is some sufficient reason for it to stop.

4. Models of Becoming for Newtonian Universes

I will start with a proposal for the sparsest version of growing block models in the Newtonian setting; if it should prove to be necessary, additional structure can be added later. To give a quick indication of what goes into the Newtonian brand of such models I will construct them by cheating, i.e. by starting with a Newtonian block model and then building a corresponding Broad model from 'chips off the old block'. Let $\mathcal{N}$ be a Newtonian block model and let $\mathcal{N}_{T\leq\Delta}$ be the future-truncated model given by deleting from the spacetime manifold of the spacetime of $\mathcal{N}$ all those points p such that $T(p) > \Delta$, $-\infty < \Delta < +\infty$, and then restricting the geometric and physical fields Gi and Pj of $\mathcal{N}$ to the truncated manifold. The intended interpretation of $\mathcal{N}_{T\leq\Delta}$ is, of course, that it represents the sum total of existence as of time $T = \Delta$. The associated Broad model is $\mathcal{B}(\mathcal{N}) := \langle\{\mathcal{N}_{T\leq\Delta} : -\infty < \Delta < +\infty\}, \precsim\rangle$. The relation $\precsim$ has the intended interpretation that $X \precsim Y$ iff Y contains at least as much existence as X. It can be defined (in cheating fashion) by the obvious condition, *viz.* $\mathcal{N}_{T\leq\Delta} \precsim \mathcal{N}_{T\leq\Delta'}$ iff $\Delta \leq \Delta'$. This relation gives a total order, since it is reflexive, transitive, antisymmetric, and connected (i.e. for any elements X and Y, either $X \precsim Y$ or $Y \precsim X$).

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Broad would surely have agreed that this way of constructing growing block models is cheating. In particular, he would say that the way in which $\precsim$ is defined above gets things metaphysically backward, since it uses the time order supplied by the absolute time function T of the block model to define an increase in the sum total of existence, whereas (according to Broad) the time order of events arises from the increase in the sum total of existence. What follows is a suggestion for constructing growing block models in a less cheating fashion that is more congenial to Broad's purposes.

Take a Broad growing block model $\mathcal{B}$ to be a pair $\langle \mathfrak{N}, \precsim \rangle$, where $\mathfrak{N}$ is a set of spacetime models, each of which is isomorphic to a chip $\mathcal{N}_{T \leq \Delta}$ off some Newtonian block model or other, and $\precsim$ is a relation on $\mathfrak{N}$ defined, as Broad would have it, in terms of the increasing of the sum total of existence: for $\mathbf{n}, \mathbf{n}' \in \mathfrak{N}$, $\mathbf{n} \precsim \mathbf{n}'$ iff $\mathbf{n}$ can be isomorphically embedded as a submodel of $\mathbf{n}'$.[11] The specification of $\mathfrak{N}$ might appear to involve an embarrassing parasitism on Newtonian block models. But the Broadheads can reply that the elements of $\mathfrak{N}$ can be specified in intrinsic geometrical terms without any reference to Newtonian block models, and the fact these elements are isomorphic to chips off block models is embarrassing only if one starts from the point of view that the block model is ontologically prior to the growing block model.[12]

The proposed construal of growing block models escapes the potential embarrassments mentioned in the preceding section (and endlessly repeated by Blockheads), since it does not make use of, or presuppose, an additional fifth dimension with respect to which the growth of the block is measured, and it renders meaningless the question of how fast the growing block is growing. ('[T]here is no sense in asking: At what speed does a certain phase, which was present, retreat into the past? And there is no sense in the suggestion that some might do it faster than others' [Broad 1959, 766].) But the flip side of this virtue of the present minimalist version of Broad Becoming models is a lack of the dynamism that advocates of a dynamical theory of time hanker after. Breaking an unanimated block model into unanimated pieces does not (they might complain) produce animation. I presume that Broad would reply that animation belongs to the province of Disney; Broad's goal is not to produce animation but to produce models that do justice to Becoming. If this can be done in a way that satisfies the cravings for dynamism, so much the better; if it can't, then so much the worse for dynamism.[13] I will not take a stand on this issue here, except to insist that providing a model for Becoming and providing a model that satisfies the hankerings for animation and dynamism are separable tasks. Another attempt to embarrass the above version of the growing block model is to ask why Becoming takes place by accreting infinitely thin layers of existence, rather than, say, temporal slabs of finite duration.[14] The only satisfactory response I can see to such questions is to propose and justify a mechanism for the birthing of events and then to show that the mechanism produces said version of Becoming. How this might be accomplished is discussed in Section 7.

That the relation $\precsim$ is defined in terms of embeddings of elements of $\mathfrak{N}$ into one another is an advantage over models of Presentism, which by analogy would be pairs $\langle \mathfrak{P}, \preccurlyeq \rangle$ where the elements of $\mathfrak{P}$ correspond to $T = const$ slices of Newtonian block models and $\preccurlyeq$ is an order relation on $\mathfrak{P}$. It seems that $\preccurlyeq$ has to be taken as an unexplained primitive. But this advantage of Broad models comes with a potential cost. As

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defined in terms of embeddability, $\precsim$ gives a *pre*order of $\mathfrak{N}$; that is, it is guaranteed to be reflexive and transitive. However, without the help of additional assumptions there is no guarantee that it is antisymmetric. Consider a temporally stationary Newtonian block model $\mathcal{N}$ (no *B*-series change).[15] Then if the elements of $\mathfrak{N}$ correspond to chips $\mathcal{N}_{T\leq\Delta}$ off this block, $\precsim$ is symmetric, since any $\mathcal{N}_{T\leq\Delta}$ can be embedded as a submodel of any other $\mathcal{N}_{T\leq\Delta'}$. Thus, it seems that the viability of the present construal of growing block models is joined at the hip with the doctrine that time requires *B*-series change, i.e. a change from-earlier-to-later in the physical fields P_j. This is not a natural alliance, however, since Broad's Becoming is supposed to provide another kind of change that is sufficient to ground temporal order and the direction of time. ("[I]f every fresh slice of existence were qualitatively indistinguishable from all its predecessors, there would still be this continual becoming' [Broad 1923, 91].) Furthermore, the present difficulty is not escaped by the alliance with the doctrine that time requires change in the behavior of the P_j. For if the elements of $\mathfrak{N}$ correspond to the chips of Newtonian block model which is non-stationary but in which the change in the P_j is strictly periodic in T, then again $\precsim$ fails to be antisymmetric.

With antisymmetry granted, the final feature of the $\precsim$-ordering needed to make it a total ordering is connectedness, and as far as I can see this feature simply has to be postulated. From the point of view of Blockheads, this postulation is a way of smuggling in the condition that the elements of N not be chips off a heterogenous collection of Newtonian block models. Broadheads can respond that there is no embarrassment here for them, since the connectedness of $\precsim$ is just a constitutive feature of the growing block model.

To rule out 'gaps' in the $\precsim$-ordering, Broadheads will want to require that the ordering be dense in $\mathfrak{N}$ and have the least upper bound property. To round out the conditions on the growing block model, Broadheads would presumably also want to require that there is no greatest element in the $\precsim$ ordering of the growing block model $\langle\mathfrak{N}, \precsim\rangle$, since otherwise the growth would come to a stop. This condition, however, is not sufficient to guarantee that the growth is maximal, i.e. that there is no $\mathbf{n}^* \notin \mathfrak{N}$ such that (a) $\mathbf{n}^*$ is isomorphic to a future-truncated block model, $\mathcal{N}_{T\leq\Delta}$ with $\Delta < +\infty$, and (b) $\mathbf{n} \precsim \mathbf{n}^*$ for every $\mathbf{n} \in \mathfrak{N}$. The Broadheads and their opponents can have the by now familiar wrangle over whether this maximality condition is a further indication of parasitism of the Broad models on block models.

Apart from the formal features of $\precsim$, there is also the worry that the Broad growing block models as presented above may leave an indeterminacy in the way the sum total of existence increases. Block models provide a criterion of trans-temporal identity for particles in the form of spacetime continuity of the particle worldlines. The basis for such a criterion is lacking in Becoming models, and as a consequence the problem of trans-temporal identity turns into the much more fraught problem of transworld identity. The point is that if $\mathbf{n} \precsim \mathbf{n}'$, i.e. n, can be embedded as a submodel of $\mathbf{n}'$, the embedding may be non-unique in disturbing ways. For ease of discussion, revert to thinking of the elements of $\mathfrak{N}$ as chips off a Newtonian block $\mathcal{N}$. Suppose that $\mathcal{N}$ describes two point particles a_n and b_n which have the same masses (and other intrinsic properties as well) and which behave symmetrically up to $T = \Delta, \Delta > 0$ (i.e. there is an automorphism

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of $\mathcal{N}_{T\leq\Delta}$ that interchanges a_{n} and b_{b}); and suppose further that after $T = \Delta$ the particles behave asymmetrically, say, because b_{n} accelerates but a_{n} doesn't (and, thus, there is no automorphism of $\mathcal{N}_{T>\Delta}$ that interchanges a_{n} and b_{n})—see Figure 1(i). Now let **n** and **n**′ be the chips $\mathcal{N}_{T\leq\Delta}$ and $\mathcal{N}_{T\leq\Delta'}$ where $\Delta' > \Delta$. Then **n** can be isomorphically embedded into **n**′ in one of two ways – either the worldlines of particles a_{n} and b_{n} of **n** are grafted, respectively, onto the worldlines of particles $a_{\mathrm{n}'}$ and $b_{\mathrm{n}'}$ of **n**′, or alternatively they are grafted, respectively, onto the worldlines of $b_{\mathrm{n}'}$ and $a_{\mathrm{n}'}$. It does no good for the Broadheads to say that because the future is unreal, there is at $T = \Delta$ no fact of the matter as to which of the particles labeled a_{n} and b_{n} in **n** will accelerate in the future. For the present version of the Broad Becomings models seems to commit them to the more radical and counterintuitive claim that even at $T > \Delta'$ there is no fact of the matter as to which of the particles a_{n} and b_{n} in **n** accelerates between $T = \Delta$ and $T = \Delta'$.

The Broadheads can respond by ruling out the above example on the grounds that the kind of Becoming they want to model is deterministic whereas the above scenario implicates indeterminism. To see why the charge of indeterminism is plausible, note that in the Newtonian setting space is isotropic so that the laws of motion must be

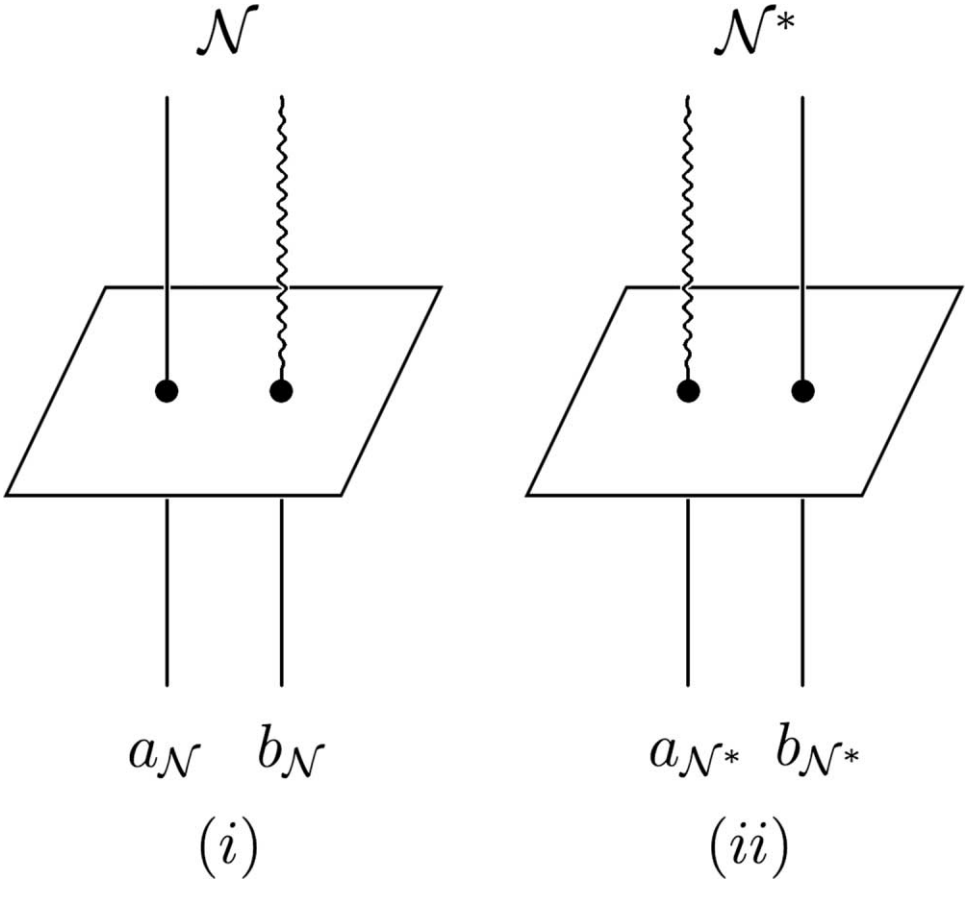


Figure 1 A Failure of Determinism (?)

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invariant under spatial rotations. Hence, if the block model $\mathcal{N}$ of Figure 1(i) satisfies the laws then so does $\mathcal{N}^*$ of Figure 1(ii), giving a violation of

> (Det) For any $\mathcal{N}$ and $\mathcal{N}^*$ satisfying the laws of motion and any $\Delta > 0$, if there is an isomorphism $I : \mathcal{N}_{T\leq\Delta} \rightarrow \mathcal{N}^*_{T\leq\Delta}$ then there is an isomorphism $\tilde{I} : \mathcal{N} \rightarrow \mathcal{N}^*$ that is an extension of I,

when I is chosen so as to match a_{n} and b_{n} to a_{n^*} and b_{n^*}, respectively. The example at issue does not violate the weaker version of (Det) which removes the requirement that $\tilde{I}$ be an extension of I, so in addition to requiring determinism the Broadheads require it in a strong form.[16] And they have to restate (Det) in terms that do not rely on block models. It is not readily apparent how the restatement would work, since the notion that $\tilde{I}$ is an extension of I relies on the fact—supplied by block models—that the domain of I is a subdomain of $\tilde{I}$.

Rather than appealing to determinism to solve the indeterminancy problem, the Broadheads might appeal to a trans-**n**-**n**′ notion of particle identity. Similarly, the above discussed problem of the antisymmetry of $\precsim$ might be solved not by appeal to the doctrine that time requires *B*-series change but by an appeal to a trans-n-n′ notion of identity of time slices.[17] But to the extent that the growing block model $\mathfrak{N}$ is distinct from a block model with a shifting 'now' that marks the progress of Becoming like a shifting bull's eye marks the progress of a policeman along a road (a notion that Broad ridicules) and is otherwise not parasitic on a Newtonian block model $\mathcal{N}$ to supply the basis of these trans-**n**-**n**′ notions of identity, then the elements of $\mathfrak{N}$ are in every relevant sense different possible worlds. Consequently, the Broadheads are not entitled to help themselves to trans-world notions of identity without some further explanation.

Now I come to what is, perhaps, the most interesting problem with the proposed version of the growing block model. To begin, note that an admissible Newtonian block model (which, recall, is required to be inextendable) is not an element of a growing block model. However, there is a sense in which block models can be regarded as ideal completions of growing block models.[18] Given all of the conditions imposed so far—that $\precsim$ provide an ordering that is total, non-gappy, and maximal—Broadheads can hope to prove that (i) for any Broad growing block model $\langle\mathfrak{N}, \precsim\rangle$ there exists a block model $\mathcal{N}$ into which each of the elements of $\mathfrak{N}$ can be embedded as a submodel; and (ii) assuming some appropriate form of determinism, this $\mathcal{N}$ is unique (up to isomorphism). If the demonstration is forthcoming, then the Broadheads can claim to have their cake and eat it too: they can reject block models as models of the universe while talking about them as ideal limits of growing block models. Of course, the Blockheads will not be overly impressed by such a demonstration, since (from their perspective) the recovery of a unique block model is an adequacy condition on a growing block model, and (they will gripe) if enough conditions are imposed on the growing block model to mimic features of the block model, then the recovery of the latter from the former is hardly surprising.

If it turns out that a growing block model does not determine a unique (up to isomorphism) block model—as is provably the case for the non-Broadian worldline Becoming models discussed below in Section 6—then Blockheads will see a strong

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confirmation of their suspicion of the unworkability of Becoming. Broadheads can react by imposing further conditions on the growing block models to achieve a uniqueness result. Or they could glory in the fact that a growing block model underdetermines a standard block model as confirming their suspicion of the mythological status of block models. Further discussion of these options will be postponed to Section 6.

Finally, it is worth reflecting on how well the proposed construal of growing block models supports Broad's doctrine that the past and present are real while the future is not. ('[T]he essence of a present event is, not that it precedes future events, but that there is quite literally nothing to which it has the relation of precedence' [Broad 1923, 66].[19]) Each element $\mathfrak{N}$ of the growing block model $\langle \mathfrak{N}, \precsim \rangle$ represents the world as it has gown up to some moment $T = \Delta$ of Newtonian absolute time, and in $\mathrm{n}_\Delta \in \mathfrak{N}$ the future of that moment, as Broad says, literally does not exist, since in n_Δ there is no spacetime to the future of that moment. But the advocates of the growing block model might seem to be forced to acknowledge the reality of the future, when accounting for the truth of tensed sentences. Suppose that in 2008 a Blockhead utters the tensed sentence *S*: 'Astronauts from earth will establish a colony on Mars'. By Blockhead standards this assertion is true iff in the block model $\mathcal{N}^@$ of the actual world there is a time $T = \Delta > 2008$ such that the untensed sentence *S'*: "Astronauts from earth establish a colony on Mars at $T = \Delta$", is true. This commits the said Blockhead to future colonies on Mars, as the asserter will readily acknowledge. Now suppose that in 2008 a Broadhead also wants to assert *S*. By Broadhead lights it is natural to take this assertion as true iff there is an element $\mathbf{n}_\Delta \in \mathfrak{N}^@$ of the growing block model $\langle \mathfrak{N}^@, \precsim \rangle$ for the actual universe such that $\Delta > 2008$ and *S'* is true in $\mathbf{n}_\Delta$. Doesn't this commit the Broadhead who in 2008 asserts *S* to future colonies on Mars? Broadheads could respond by withdrawing the assertion or by trying to find a different semantics that makes the assertion true while avoiding the appearance of a commitment to the reality of the future. But I think that there is no need to do either. The advocates of the growing block model should concede that they cannot coherently sustain an absolute denial of the reality of the future; that would require giving up the growth in the growing block model in favor of some one fixed future-truncated block model. For the growing block model the denial of the reality of the future can only be perspectival, e.g. from the perspective of the world n_{2008} as it has grown up to $T = 2008$, future times are unreal. Of course, Blockheads who operate with a future-inextendible block model can also say that from the perspective of $T = 2008$ future times are unreal, but this denial of the reality of the future amounts to no more than the triviality that the future does not exist at the present time. Even in advance of deciding the best way to work the semantics of tensed sentences for growing block models, it is safe to say that advocates of the growing block models can underscore the distinctiveness of their version of the unreality of the future by holding to the following claims: 'The sufficiently distant future is literally unreal/literally does not exist (i.e. there is no spacetime sufficiently far to the future of now)' and 'The sum total of existence was smaller and will be larger'; for such assertions are true *simpliciter* for the growing block model $\mathfrak{N}^@$ of the actual universe, i.e. true in every $\mathbf{n} \in \mathfrak{N}^@$. Moreover, these claims are necessarily true, since they are true *simpliciter* in every growing block model $\mathfrak{N}$.

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I think a fair summary of the discussion thus far is that in the setting of Newtonian spacetime—the setting Broad had in mind—a number of substantive conditions (the prohibition against stationary models and strictly periodic models, the apparent need to rely on a strong form of determinism) and/or conditions that arouse suspicions of parasitism on block models are needed to produce a version of the growing block model that is suitable to Broad's metaphysics of Becoming and that also satisfies seemingly natural demands on models of time and change. The need for such conditions does not show that the growing block model is unworkable, but it does reduce its attractiveness and, thus, gives encouragement to the Blockhead programme of trying to show how block models can accommodate all that can be legitimately claimed about the nature of change. Outside the Newtonian setting things get worse for the growing block model.

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5. Models for Hypersurface Becoming for Relativistic Universes

It might seem that any discussion of whether Broad's notion of Becoming can be transferred to the relativistic setting should begin by addressing the much discussed arguments of Putnam (1967) and Rietdijk (1966), which purport to show that the view that the future is unreal is incompatible with STR. The assumptions of these arguments have been challenged (see, for example, Stein 1991), but the main point to emphasize here is that these arguments are oblique to Broad's project. Putnam and his critics alike assume a block spacetime model—in fact, Minkowski spacetime, but it doesn't matter for the general point I wish to make which (future-inextendible) block spacetime model is chosen. Then they debate about the appropriate way of attaching labels ('real' or 'exists') to events in the model. One way to score debating points is to show that some allegedly natural base conventions about the assignment of labels, together with facts about the structure of spacetimes, force an assignment of labels to future events—e.g. Putnam argues that an observer here–now in Minkowski spacetime must, on pain of inconsistency with his base conventions, assign the label 'real' to events in the future light cone of here–now. For someone interested in ontology it is hard to get excited about such debates, since all the participants agree on the spacetime structure and on the events that populate this structure. But Broad's view of the unreality of the future (at least in *Scientific Thought*) has nothing to do with the labeling of events or parsing the meaning of 'is' or 'exists'; rather, in denying the reality of the future he is putting forward an ontological thesis in which there literally is no future in the sense made clear by future-truncated block models. (To repeat, '[T]he essence of a present event is, not that it precedes future events, but that there is quite literally nothing to which it has the relation of precedence' [Broad 1923, 66].)

Of course, one of the key considerations in the Putnam–Rietdijk arguments—the absence of an absolute simultaneity in the standard version of STR—does raise concerns about how Broad's models of Becoming can be modified to accommodate relativistic spacetime structure. This is an issue that will be addressed in this and the following section. But at the outset it should be emphasized as strongly as possible that *neither STR nor GTR* per se *is inconsistent with Broad's view—in particular, neither*

theory is inconsistent with future-truncated spacetime models, although it is possible to construct relativistic laws that would be inconsistent with such truncations, as I will explain presently.

It will be assumed for sake of simplicity that all of the relativistic spacetimes M, g_{ab} of the relativistic block models to be discussed here are time orientable and that a time orientation has been singled out.[20] Thus, relations of *chronological precedence* $\ll$ and *causal precedence* $<$ can be defined, respectively, as follows: for $p, q \in M$, $p \ll q$ iff there is a future directed timelike curve from p to q, and $p < q$ iff there is a future directed non-spacelike curve from p to q. The *chronological past* $I^-(p)$ (respectively, chronological future $I^+(p)$) of a point $p \in M$ is the set of all points $q \in M$ such that $q \ll p$ (respectively, $p \ll q$). The *causal past* $J^-(p)$ and the causal future $J^+(p)$ of a point $p \in M$ are defined analogously. The relation of chronological precedence can in turn be used to define the notion of a *global time function* as a smooth map $t : M \to \mathbb{R}$ such that for p, $q \in M$ with $p \ll q$, $t(p) < t(q)$.[21]

For relativistic block model $\mathcal{R}$ whose spacetime admits a global time function t, the cheating construction of the relativistic version of Broad-type growing block models can ape the cheating construction of the Newtonian version of the growing block models by using the global time function t in place of Newtonian absolute time T. The associated Broad Becoming model $\mathcal{B}(\mathcal{R}, t)$ will have the form $\langle \{\mathcal{R}_{t \leq \delta} : l < \delta < u\}; \precsim \rangle$, where $\mathcal{R}_{t \leq \delta}$ is the future-truncated model given by deleting from the spacetime manifold of $\mathcal{R}$ all those points p such that $t(p) > \delta$ and restricting the geometric and matter fields of $\mathcal{R}$ to the truncated spacetime; l and u indicate the lower and upper bounds on the range of t; and $\precsim$ is defined in the obvious way, i.e. $\mathcal{R}_{t<\delta} \precsim \mathcal{R}_{t \leq \delta'}$ iff $\delta \geq \delta'$.

Most of the qualms about the non-cheating Newtonian version of growing block models carry over to attempts to formulate a non-cheating relativistic version, and they need not be repeated here. One exception concerns the worry about the desired antisymmetry property of the relation $\precsim$ as defined by embeddability. While this worry does not completely disappear in the case of GTR, it is considerably ameliorated. The counterexamples to the antisymmetry of $\precsim$ for the non-cheating version of the Newtonian growing block models relied on the fact that Newtonian time is infinite in the past and on the existence of physically possible scenarios either temporally stationary or strictly periodic. But only rather exceptional and artificial solutions to Einstein's field equations for GTR are temporally stationary or else non-stationary but strictly periodic in time,[22] and cosmological solutions that are viable candidates for models of the actual universe are not infinite in the past.

But with this one exception, things only get worse for the Broadheads when they try to implement the growing block model in a relativistic setting. Whether or not Broad's metaphysics of Becoming can be overlaid on a relativistic theory without inducing new empirical consequences or, worse, rendering the theory incoherent depends on substantive features of the theory—an obvious point that seems to have escaped commentators. Relativistic theories whose laws are stated in terms of local partial differential equations (e.g. Maxwell's theory of electromagnetism set in Minkowski spacetime and Einstein's GTR) are tame in this regard, for such laws have the property that if a relativistic block model $\mathcal{R}$ satisfies them, then so does any future-truncated

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model $\mathcal{R}_{t\leq\delta}$, $\delta < u$. This property is not shared by all relativistic theories. Consider the class of theories, initially intended to be implemented in Minkowski spacetime, which offer Lorentz invariant equations of motion for particles acting at-a-distance (i.e. without intervening fields). In one subclass of such theories the force exerted by a particle *j* on another particle *k* at some point *p* on *k*'s worldline depends on the velocity and acceleration of *j* at those points where *j*'s worldline intersects the past and future light cone of *p*.[23] In this subclass the application of the laws of motion to $\mathcal{R}$ and to the future-truncated model $\mathcal{R}_{t\leq\delta}$ may produce different verdicts on what motions in $\mathcal{R}_{t\leq\delta}$ are physically possible. And worse, the application to future-truncated models can lead to a reversal of verdicts as the growing block grows, i.e. the motions described in $\mathcal{R}_{t\leq\delta}$ may satisfy the laws of motion while the prolongations of these motions as described in $\mathcal{R}_{t\leq\delta'}$, $\delta' > \delta$, do not, or vice versa. The Broadheads will, of course, remain unphaged by such results; indeed, they will see such results as confirmation of what they already know; namely, since the future is unreal, any physical theory that attempts to embody a lawlike dependence of past or current events on the nonexistent future is bound to lead to incoherence. But by the same token, the Broadheads expose themselves to empirical disconfirmation should the theories at issue prove to be empirically accurate when applied in the originally intended manner to Minkowski block models.

I turn now to a dilemma that the Broadheads face in constructing their growing block models in the general relativistic setting. Some general relativistic spacetimes belonging to solutions of the field equations of GTR do not admit a global time function, while those that admit one global time function admit an uncountable infinity of them. In the first instance, the construction of growing block models cannot ape the construction for Newtonian spacetime. In the latter instance, the embarrassment of riches of time functions leads to the embarrassing question of which time function corresponds to the growth of the growing block.

To sharpen the first horn of the dilemma, note that the necessary and sufficient condition for a general relativistic spacetime *M*, g_{ab} to admit a global time function is *stable causality*, which requires that there exist a non-vanishing timelike vector field t^a such that the spacetime *M*, $\tilde{g}_{ab}$, where $\tilde{g}_{ab} := g_{ab} - t_a t_b$, contains no closed timelike curves (intuitively stated, there exists a finite widening of the null of cones of the metric g_{ab} that does not result in closed timelike curves). Stable causality is not only stronger than the requirement of *chronology*, i.e. that the spacetime not admit any closed future-directed timelike curves, it is also stronger than the requirement that the spacetime not admit any almost closed timelike curves (a.k.a. *strong causality*).[24] There are attempts in the literature to show that the existence of closed timelike curves would produce antinomies (e.g. the 'grandfather paradox') and, thus, that spacetimes containing closed timelike curves are physically impossible. I am persuaded that such a conclusion is unjustified, but I will not rehearse the reasons here.[25] But the point to emphasize here is that even if such a conclusion is warranted, it does not extend to a justification for classifying non-stably causal spacetimes as physically impossible, since, as far as I am aware, there are no antinomies comparable to the grandfather paradox for non-stably causal but strongly causal spacetimes.

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The Broadheads can respond in various ways to the first horn of the dilemma. (R1) Broadheads can claim that non-stably causal spacetimes are metaphysically impossible because necessarily Becoming takes place and necessarily it takes place by accreting successive layers of existence that correspond to the level surfaces $t = const$ of a global time function for the block model that is the ideal completion of the growing block model. Blockheads will, of course, reject the first necessity claim. And broad-minded Broadheads may also wish to reject the second necessity claim and pursue a different response. (R2) Broadheads can switch to a different form of Becoming where the accretion of existence does not consist of adding layers of existence corresponding to the level surfaces of a time function and that does not require that the ideal completion of the growing block model yield a block spacetime satisfying stable causality. A version of this response will be studied in the following section. (R3) Broadheads can respond more modestly that the Becoming story cannot be told for a block model whose spacetime is non-causally stable. But, they could continue, they do not intend their story to hold for every possible world but only for the actual world and (perhaps) for nearby possible worlds. The viability of this response depends on the assumption that all of the elements of the growing block model for the actual universe, as well as the block spacetime that is the ideal completion of the growing block model, are stably causal. Empirical evidence on this matter is hard to marshal, since stable causality is a global property of the spacetime, and as such the presence or absence of the property may not be decidable by the observations obtainable by any possible observer.[26]

The second (embarrassment-of-riches) horn of the dilemma can be dealt with in various ways. One way would be to single out one of the plethora of global time functions and declare that Becoming proceeds by adding layers of existence corresponding to the level surfaces of this function. Two problems arise here. In some general relativistic block models it may be possible to identify one of the global time functions as geometrically or physically natural. For example, matter may be in the form of a perfect fluid with four-velocity V^a, and V^a may be orthogonal to the level surfaces $t = const$ of some global time function t.[27] But this method of privileging does not work across the board, because it is known that there are models satisfying the Einstein field equations of GTR where the velocity field V^a of the perfect fluid is not hypersurface orthogonal.[28] Nor does this method of privileging guarantee uniqueness, because there are solutions of the Einstein field equations where there are two or more independent fluids with different velocity fields orthogonal to different families of spacelike hypersurfaces. Another method of privileging is to use a time function, invariantly defined in terms of the spacetime geometry, corresponding to a foliation of spacelike hypersurfaces of constant mean curvature. Typically such a foliation is unique when it exists, but existence is guaranteed only for a limited class of solutions to Einstein's field equations, a class that does not exhaust solutions with causally nice features (see Rendall 1996, for an overview). Other methods of privileging could be considered, but there does not seem to be one that is physically well-motivated and that singles out a unique time function for all the general relativistic spacetimes for which one would hope Becoming would apply.

The second and more metaphysically challenging problem is to show why Becoming should track one time function rather than another. A partial answer to this challenge,

and to the first challenge as well, might come from the notion that Becoming should take place in a deterministic fashion. (In Section 3 we saw a reason for wanting determinism when constructing Newtonian Becoming models.) In terms of block relativistic models this constraint implies that the time function that Becoming tracks is one whose level surfaces are *Cauchy surfaces* (i.e. spacelike hypersurfaces each of which is intersected exactly once by every endless non-spacelike curve), which are the appropriate launching pads for the relativistic version of Laplacian determinism (see Earman 2007). But this only partially answers the challenge, since, although many time functions are ruled out, it is still the case that if there is one Cauchy time function there are infinitely many. And the demand for deterministic Becoming limits the applicability to general relativistic spacetimes that are *globally hyperbolic*, a condition even stronger than stable causality.[29]

A perhaps more promising way of dealing with the embarrassment-of-riches horn is to provide a mechanism generating an increase in the sum total of existence and then to show that the increase corresponds to the accretion of the layers of 'now' corresponding to a particular time function. The mechanism could come either from theorizing in physics or from armchair philosophizing. As an example of the former, the causal set approach to quantum gravity (to be discussed below in Section 7) studies a mechanism for a stochastic 'birthing' of events; but the resulting way in which the sum total of existence increases may not conform even approximately to a Broad hypersurface Becoming model.

Tooley's *Time, Tense, and Causation* (1997) is an example of the armchair. Tooley begins with a transcendental deduction: find what commitments lie behind our judgments about time, change, and causation; and infer that the world must have features that make good on these commitments. Tooley infers that events in our world are causally related, that the cause–effect relation must be asymmetric, and that causes produce their effects by bringing them into being. This view of causation is joined with a substantivalist view of spacetime according to which spacetime is composed of spacetime points, which are contingent entities. Each spacetime point causes other spacetime points to exist, and was itself caused to exist by other preexisting points. Tooley postulates that in the context of STR the causally related spacetime points lie on a timelike geodesic of Minkowski spacetime and that the geodesics corresponding to different causally related chains of spacetime points are parallel. This timelike geodesic congruence corresponds to a special inertial frame—the Absolute Frame. The advancing front of existence for Becoming in the special relativistic setting coincides (for reasons that escape me) with the spacelike hypersurfaces of the natural simultaneity of the Absolute Frame (i.e. the hypersurfaces orthogonal to the geodesic congruence). The existence of the Absolute Frame, Tooley argues, is not only not incompatible with STR but also provides a version of the theory superior to the standard one, in that it is not committed to the unverifiable proposition that the one-way speed of light is the same in all inertial frames. And, moreover, Tooley holds that his version of STR helps to rationalize the results of the actual experiments of Aspect and others that implement the Einstein–Rosen–Podolsky thought experiment; for he takes the outcomes of these experiments to show that if quantum mechanics is true, then the

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standard version of STR is incomplete, and the needed completion is the Absolute Simultaneity provided by the Absolute Frame.

While I marvel at Tooley's *tour de force* of meta-physics and metaphysics, I disagree with it at every turn. Spelling out the details of the disagreement would require a book at least as long as Tooley's own, and I will content myself here with only one observation.[30] The world we live in is described not by STR but by GTR. The diffeomorphism invariance of GTR is almost universally interpreted by physicists (and right-thinking philosophers of physics) as a gauge symmetry, i.e. as relating different descriptions of the same physical situation.[31] This interpretation tends to undermine the spacetime substantivalism essential to Tooley's account.[32] Of course, GTR is hardly the final word of physics. The Holy Grail of twenty-first-century physics is a quantum theory of gravity that combines the insights of GTR with those of quantum physics. As already mentioned, one of the extant approaches to quantum gravity (described in Section 7) does offer support for some of Broad's ideas about Becoming, but none of the extant approaches of which I am aware can happily accommodate Tooley's story.

A final and more radical way of dealing with the embarrassment-of-riches horn is to relativize hypersurface Becoming to a time function. Gödel (1949) would have objected that such a relativization destroys the objectivity of Becoming, which he took to be an essential feature of it.[33] But broad-minded Broadheads can part company with Gödel and adopt a Becoming model of the world that is, not a simple Broad model $\mathcal{B}(\mathcal{R}, t)$ but a *super Broad hypersurface Becoming model* consisting of the set of Broad hypersurface Becoming models, one for every global time function (or every time function belonging to some privileged set of the time functions) of the spacetime of $\mathcal{R}$. Objectivity of Becoming is maintained in the sense that the different elements $\mathcal{B}(\mathcal{R}, t)$, $\mathcal{B}(\mathcal{R}, t')$, $\mathcal{B}(\mathcal{R}, t'')$, ... of the super Broad model represent different aspects of Becoming in the *same* world, in that the ideal completions of each of these elements produces the same block model.

Of course, the guarantee of this minimal objectivity is a consequence of the cheating nature of the construction, which starts with a block model. A non-cheating construction—should such be produced—could guarantee this objectivity by theft or honest toil. The theft method is to distinguish between acceptable and unacceptable super Broad models and to stipulate that the satisfaction of the minimal objectivity condition is necessary for acceptability. This will bring a chortle from the Blockheads who get for free what the Broadheads achieve by fiat.[34] The method of honest toil would presumably proceed by adding some additional structure to the non-cheating version of the super Broad model that would link together and guarantee the coherence of the elements that serve as the non-cheating surrogates for the $\mathcal{B}(\mathcal{R}, t)$, $\mathcal{B}(\mathcal{R}, t')$, $\mathcal{B}(\mathcal{R}, t'')$, ... of the cheating version of the model. But it is hard to see how these linking conditions would work, since the truncated models $\mathcal{R}_{t\leq\delta}$ obtained by using different time functions can be very different; in particular, there may be no simple relation of embeddibility between them.

In sum, seizing either horn of the dilemma leads into a thicket of problems. The Broadheads may be able to beat their way out of the thicket, but after exhausting themselves on bush beating, they may wish to opt for ditching hypersurface Becoming in favor of another form of Becoming.

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6. Worldline Becoming Models in Relativistic Spacetimes

There are at least two motivations for considering worldline Becoming in relativistic spacetimes: it requires spacetimes satisfying a causality condition weaker than the condition of stable causality required by hypersurface Becoming, and it is more closely related to the way observers experience the coming into being (or, at least, the coming into purview) of events. Additionally, the results of Clifton and Hogarth (1995) could be used to motivate the position that for an inertial observer in Minkowski spacetime, what has Become as of some point on the observer's worldline consists of the events in the causal past of that point. However, the discussion here is intended to apply not only to Minkowski spacetime but to general relativistic spacetimes as well, and once the symmetries of Minkowski spacetime are lost, the Clifton-Hogarth results on the unique definability of the Becoming relation no longer apply. In any case, the goal here is not to achieve unique definability results but rather to explore a notion of Becoming that seems natural in the relativistic setting and avoids some of the difficulties attending hypersurface Becoming.

A mixed-mode expression of the idea of worldline Becoming would have it that, for an observer, the sum total of existence increases with the increase in the volume of the observer's past light cone as the observer's 'now' moves up her worldline. Of course, Broadheads who wish to adopt this mode of Becoming must show how to explicate the underlying idea without resort to McTaggartian transitory 'now' or an underlying block model. The first (but not the second) demand can be met in the now familiar cheating fashion by starting with a relativistic block model $\mathcal{R}$ and using appropriate chips off this block to construct a growing block model. An idealized immortal observer is represented by a past and future endless worldline γ of the spacetime of $\mathcal{R}$.[35] The observer-relativized worldline Becoming model $\mathcal{B}(\mathcal{R}, \gamma)$ based on $(\mathcal{R}, \gamma)$ is then $\langle \{\mathcal{R}_{J^-(p)} : p \in \gamma\}, \precsim \rangle$, where $\mathcal{R}_{J^-(p)}$ is the model obtained by deleting all of the points of the spacetime $\mathcal{R}$ not in $J^-(p)$ and restricting the geometric and matter fields of $\mathcal{R}$ to $J^-(p)$; and $\precsim$ is the relation defined by $\mathcal{R}_{J^-(p)} \precsim \mathcal{R}_{J^-(p)}$ iff $J^-(q) \subseteq J^-(r)$.

In order that increasing existence for the observer with worldline γ coincide with increasing chronological order along γ, the spacetime of the block model $\mathcal{R}$ used as the basis of the construction must satisfy the condition that if $q, r \in \gamma$ and $q \ll r$, then $J^-(q)$ is a proper subset of $J^-(r)$. And if the construction is to work for an arbitrary worldline, the condition becomes that for any q and r such that $q \ll r$, $J^-(q) \subset J^-(r)$. This condition is entailed by the causal-past distinguishing condition

$$\text{(CaPD)} \quad J^-(q) = J^-(r) \Rightarrow q = r$$

which in turn is entailed by the chronological-past distinguishing condition

$$\text{(ChPD)} \quad I^-(q) = I^-(r) \Rightarrow q = r$$

The relationship between these conditions and the other causality conditions mentioned above is given by the following chain of implications, each of which is strict:

$$\text{global hyperbolicity} \Rightarrow \text{stable causality} \Rightarrow \text{strong causality} \Rightarrow$$
$$\text{ChPD} \Rightarrow \text{CaPD} \Rightarrow \text{chronology}$$

The Broadheads must still tell some story about what to do about non-CaPD spacetimes, but since the CaPD condition is well below stable causality in the causality hierarchy, their task is much easier than in the case of hypersurface Becoming where they have to tell a story about what to do about non-stably causal spacetimes.

So far I have only introduced (in a cheating way) the observer-relativized worldline Becoming model $\mathcal{B}(\mathcal{R}, \gamma)$. A super worldline Becoming model $\mathfrak{P}$ would be specified (again in a cheating way) as a collection of $\mathcal{B}(\mathcal{R}, \gamma)$'s with $\mathcal{R}$ fixed and γ varying: in the case of democracy of observers, γ ranges over all the past and future endless worldlines of the spacetime of $\mathcal{R}$; in the case of privileged observers, γ ranges only over the past and future endless worldlines satisfying some privileging condition (e.g. γ is a timelike geodesic). Of course, honest Broadheads would want to redo the construction without using chips off a relativistic block model. There are by now familiar problems that arise for attempts at non-cheating constructions, but I will waive them in order to concentrate on one fundamental problem that threatens to undermine the viability of worldline Becoming, at least from the Blockhead perspective. Broadheads will try to turn it to their advantage.

Suppose that a super worldline Becoming model $\mathfrak{P}$ has been specified by some non-cheating construction. To ensure the objectivity of Becoming, the elements of $\mathfrak{P}$ that are the surrogates of the $\mathcal{B}(\mathcal{R}, \gamma)$ in the non-cheating construction must fit together in some appropriate way. Blockheads will insist that, at a minimum, this means that the surrogates are all embeddible into the same relativistic block model. Blockheads will also think it natural to require that the elements of $\mathfrak{P}$ should collectively fix some unique (up to isomorphism) relativistic block model. To discuss the prospects for satisfying this latter constraint, grant for charity's sake that each element of $\mathfrak{P}$ can be represented (in some natural sense) as $\langle \{\mathcal{R}_{J^-(p)} : p \in \gamma\}, \precsim \rangle$ for some relativistic block model $\mathcal{R}$ and some worldline γ of the spacetime M, g_{ab} of $\mathcal{R}$. And suppose that the total Becoming for γ, i.e. $J^-(\gamma) := \cup_{p \in \gamma} J^-(p)$, is unique up to isomorphism; that is, if $\langle \{\mathcal{R}'_{J^-(p)} : p \in \gamma'\}, \precsim \rangle$ is another representation, then $J^-(\gamma) = \cup_{p \in \gamma} J^-(p)$ and $J^-(\gamma') = \cup_{q \in \gamma'} J^-(q)$ are isomorphic. If $J^-(\gamma) = M$ then there is a natural sense in which this element of $\mathfrak{P}$ at issue fixes a block model. Then the Broadheads can stipulate that in an acceptable super worldline Becoming model, all the elements fix the same block model. But even in the most plain vanilla relativistic spacetimes, e.g. Minkowski spacetime, there are future endless timelike curves γ whose total Becoming $J^-(\gamma)$ is a proper subset of the spacetime. Figure 2 shows two-dimensional Minkowski spacetime and the worldline $\hat{\gamma}$ of an observer who undergoes hyperbolic acceleration; in this case $J^-(\hat{\gamma})$ is confined to the region below the null line L. Restricting attention to a privileged set of observers, say, those whose worldlines are geodesics, will overcome this problem – for Minkowski spacetime. But general relativistic spacetimes can exhibit the phenomenon of event horizons where for every timelike γ, $J^-(\gamma)$ fails to exhaust the entire spacetime.[36] For such spacetimes, concordance cannot be secured by stipulation. A concrete example is given by de Sitter spacetime.[37] This

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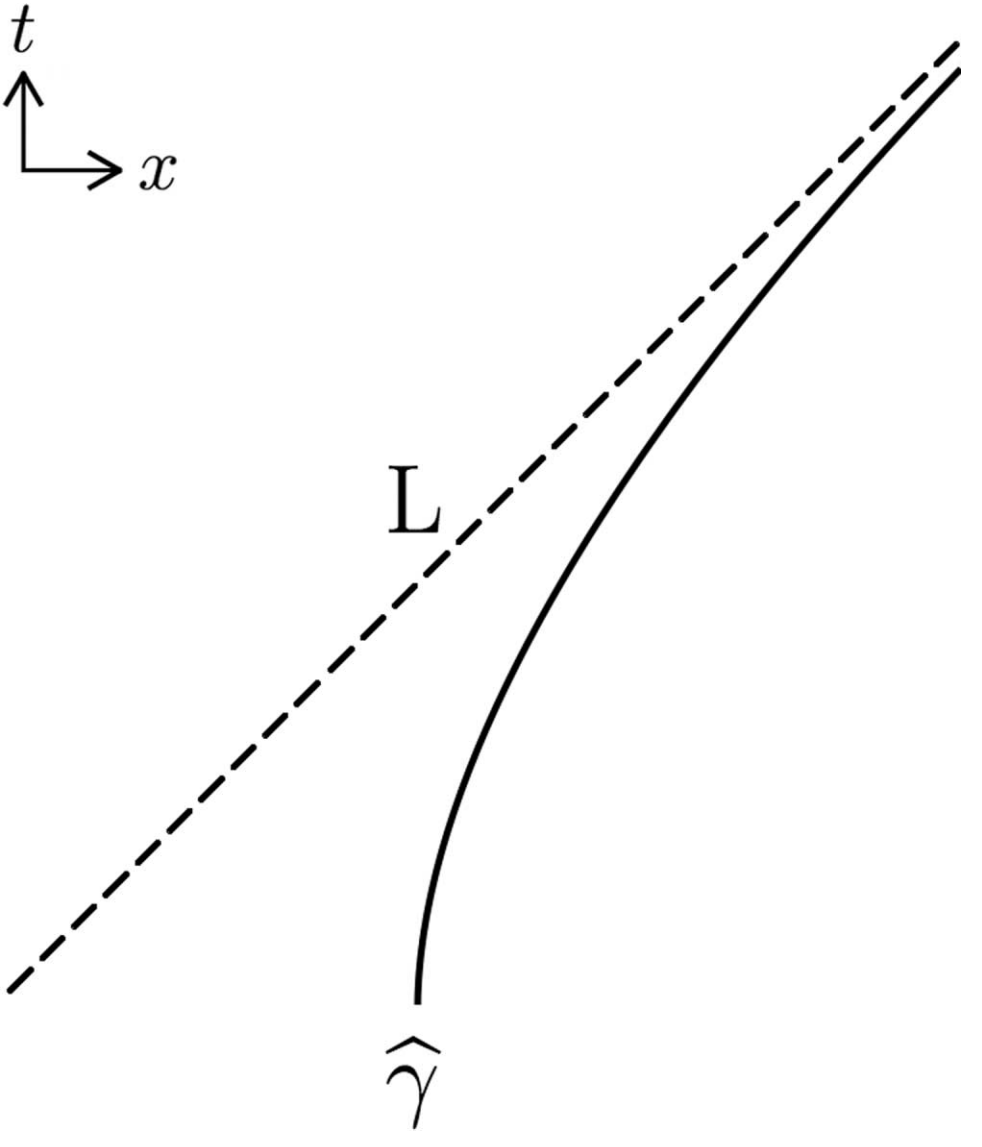


Figure 2 A Particle Undergoing Uniform Acceleration in Minkowski Spacetime.

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example takes on more than mathematical curiosity, following the recent discovery of the accelerated expansion of the universe.[38] If it should turn out that the acceleration is due to a positive cosmological constant, then the future asymptotic structure of our spacetime is that of de Sitter spacetime.

Nevertheless, it might be hoped that, although the individual elements of $\mathfrak{P}$ fail to fix a unique (up to isomorphism) block model, collectively they do achieve fixation. But the phenomenon of observational indistinguishability of general relativistic spacetimes dashes this hope. Two spacetimes M, g_{ab} and M', g'_{ab} are said to be *observationally indistinguishable* iff for any future endless worldline γ of M, g_{ab} there is a future endless

worldline γ' of $'$, g'_{ab} such that $J(\gamma)$ and $J(\gamma')$ are isomorphic, and vice versa with M, g_{ab} and M', g'_{ab} interchanged.[39] Malament (1977) shows how to construct distinct but observationally indistinguishable spacetimes, and also shows that various global properties of relativistic spacetimes—stable causality being among them—are observationally undecidable because one of a pair of observationally indistinguishable spacetimes may have the property while the other lacks it. In addition, Malament notes that the definition of observational indistinguishability for spacetimes is easily extended to a definition of observational indistinguishability of general relativistic models that include matter fields: $\langle M, g_{ab}, P_1, P_2, \ldots\rangle$ and $\langle M', g'_{ab}, P'_1, P'_2, \ldots \ldots\rangle$ are deemed to be observationally indistinguishable iff their spacetimes are observationally indistinguishable and the isomorphisms between the total Becoming sets preserve the values of the matter fields, i.e. for any such isomorphism φ, $\varphi^* P_i = P'_i$ for all i. One can then proceed to construct examples of such observationally indistinguishable models that satisfy Einstein's gravitational field equations.

Within the Blockhead and Broadhead camps the phenomenon of observationally indistinguishable spacetimes has different resonances. For the Blockheads this phenomenon is just a species of the familiar genus of epistemic underdetermination of theory and/or models of theories by some class of empirical evidence. For them it is not a problem but a fact of life in relativistic spacetimes: evidence that even an immortal observer could acquire by causal contact with the cosmos may not suffice to determine various large-scale features of it. This is an interesting and, perhaps, depressing fact, but its discovery is hardly earthshaking, since similar facts about epistemic underdetermination are familiar fare in philosophy of science.

For the Broadheads the phenomenon at issue initially poses a problem about the coherence of the collection of observer-relativized Becomings forming the Becoming model $\mathfrak{B}$. The Broadheads may complain that they have been saddled with an overly strong coherence condition; it suffices, they may contend, that the observer-relativized Becomings can be jointly realized in some block relativistic model. That all the observer-relativized Becomings can be jointly realized in distinct block models is actually an advantage because it punctures the pretense of block models. From the Broadhead perspective, to say that the cosmos is as represented by the block model $\langle M, g_{ab}, P_1, P_2, \ldots\rangle$ rather than by the block model $\langle M', g'_{ab}, P'_1, P'_2, \ldots\rangle$ can mean nothing more than that the former emerges as the unique ideal completion of the Becoming model. It follows that if uniqueness fails, then there simply is no fact of the matter about whether the cosmos is as represented by $\langle M, g_{ab}, P_1, P_2, \ldots\rangle$ as opposed to $\langle M', g'_{ab}, P'_1, P'_2, \ldots\rangle$ Hence, for the Broadheads it is not a matter of epistemic underdetermination but rather of ontic underdetermination; and the latter redounds to the discredit of the Blockheads, since they are committed to there being a fact of the matter, albeit one not decidable by observational evidence, where (according to the Broadheads) there is none.

The Blockheads may retort that this bit of hocus pocus should be seen as a *reductio* of observer-relativized Becoming, since obviously there is a fact of the matter as to whether the global structure of the cosmos is of one type or of another. At this juncture one may wish to disengage from the Broadhead–Blockhead exchange because it has degenerated into a shouting match.

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7. Becoming and the Causal Set Approach to Quantum Gravity

The search for a quantum theory of gravity is currently dominated by two research programmes, string theory and loop quantum gravity. A third programme, called the causal set approach (Bombelli et al. 1987; Reid 2001), is much further away from qualifying as a fully fledged quantum theory of gravity. But it will be discussed here because it promises to illuminate, if not resolve, some of the key problems encountered in trying to make Broad's Becoming work in a relativistic spacetime setting.

A *causal set C* (or *causet* for short) is a set equipped with a relation $\prec$ that is transitive, irreflexive, and locally finite in the sense that for any $e, e' \in C$, $Card(Past(e) \cap Fut(e')) < \infty$, where $Past(e) := \{f \in C : f \prec e\}$ and $Fut(e') := \{f' \in C : e' \prec f'\}$; and $Card(X)$ is the cardinality of the set X. The relation between causets and the spacetime models of classical GTR is conceived as one of representation and approximation. Call a map $\phi : C \rightarrow M$, g_{ab} a faithful embedding of the causal set C into a relativistic spacetime M, g_{ab} iff (i) $\prec$ is represented as causal precedence, i.e. for all $e, e' \in C$, if $e \prec e'$ then $\phi(e) \in J^-(\phi(e'))$ and (ii) the set of points $\phi(C) \subset M$ is a high-probability set whose elements are sprinkled randomly at a density ρ by a Poisson process. Such an embedding may not exist, but if it does, it has been conjectured (Bombelli et al. 1987) that the embedding is essentially unique in that if $\phi' : C \rightarrow M'$, g'_{ab} is another faithful embedding, then there is a diffeomorphism $d : M \rightarrow M'$, such that $\phi' = d \circ \phi$ and $d^* g'_{ab} \approx g'_{ab}$, where the approximation is judged on the discreteness scale ρ^{-1}.

Causets can grow by 'birthing' new elements. The advocates of the causal set approach are explicit in linking this growth to the passage of time. Thus, Rideout and Sorkin (1999) write that 'The phenomenological passage of time is taken as a manifestation of this continuing growth of the causet. Thus, we do not think of this process as happening "in time", but rather as "constituting time"' (024002-2). The name of C. D. Broad is not invoked in this *Physical Review* article, but the similarities of the birthing process to Broad's notion of Becoming are explicitly acknowledged by Sorkin (2007) in an article entitled 'Relativity Theory Does Not Imply That the Future Already Exists,' a title that would surely have warmed Broad's heart.

The significantly new aspect added by the advocates of causal sets is a dynamics by which the growth in the causets takes place. In the simplest dynamics, called the *classical sequential growth dynamics* (Rideout and Sorkin 1999; Varadarajan and Rideout 2006), the birthing is a discrete-stage, stochastic process starting from the empty set and adding elements one by one, with the transitions governed by probabilities satisfying a Markov condition.[40] As the name indicates, this dynamics is classical in that it does not try to capture quantum interference effects, and consequently it foregoes the possibility of modeling quantum gravity effects involving interference. And even at the classical level it has the drawback of not producing manifold-like causets (see Georgiou 2005). However, for present purposes the (over) simplified nature of the dynamics has the advantage of allowing key conceptual points relevant to Becoming to stand out.

One natural way to model the causet version of Becoming in this setting is by means of a set C_∞ of finite causets that is totally ordered by the parent–child relation $\rightsquigarrow$ and that has a first element consisting of the empty set but has no last element.

Here $C \rightsquigarrow C'$ is an allowed (non-zero probability) transition from an n-element causet C to an $(n + 1)$-element causet $C' = C \cup \{d\}$, with d the 'daughter' of the 'parent' C.[41] Given the intended interpretation of the model, it would be embarrassing if the daughter d stood in the relation $d \prec e$ to any element e of the parent C, since this would mean that events could be born in the past of events that have already Become. The embarrassment could be avoided by stipulation; in the causets literature this stipulation is called the requirement of *internal temporality*.

This very brief sketch should be sufficient to make it apparent that the causal set approach has the resources to deflect the embarrassing questions confronting the Broad models of Becoming in both the Newtonian and relativistic contexts: Why does existence increase by accreting infinitely thin layers rather than finite slabs or some other configuration? Assuming that existence does increase by accreting infinitely thin layers, why should these layers correspond to the level surfaces of one particular cosmic time function rather than another? The causets answer to both is a *laisse-faire* conception of Becoming that doesn't try to shoehorn Becoming into an *a priori* mould. Existence increases by the birthing of events through a discrete stochastic process. When the results of this process are approximated by a sequence of classical general relativistic spacetimes, the result may resemble a hypersurface Becoming model of the type discussed in Section 5, or it may not—that depends on the outcomes of the stochastic process.

Still, it is natural to ask about the prospects that causet Becoming will produce something resembling hypersurface Becoming. Towards this end, define an anti-chain $A \subset C$ of a causet C to consists of elements no two of which are $\prec$-related. A is a maximal anti-chain for C if any $c \in C$ that is not in A is $\prec$-related to some element of A. The maximal elements of a (finite) causet C form a maximal anti-chain F_C that may be thought of as the 'future boundary' of C. Supposing that C admits a faithful embedding $\phi : C \rightarrow M, g_{ab}$, is it the case that F_C is represented by a time slice in the sense that $\phi(F_C)$ is well approximated on the appropriate discreetness scale by a slice of M, g_{ab}? In general a faithful embedding of a causet does not represent an anti-chain as a time slice, but when the embedding spacetime is spatially open there are indications that, with high probability, it does represent an anti-chain as a time slice (Major et al. 2006). Suppose next that the causet Becoming model C_∞ has the property that sufficiently far along in the parent–child chain every successor child $C \in C_\infty$ admits a faithful embedding ϕ into the same general relativistic spacetime M, g_{ab}, that this embedding is essentially unique, and that $\phi(F_C)$ is well approximated by a time slice of the embedding spacetime. Does it follow that these time slices all correspond to the level surfaces of the same global time function of M, g_{ab}? It seems reasonable to suppose so, but I know of no proof. Only if all of these suppositions hold for what are in some appropriate sense generic causet Becoming models can it be said that hypersurface Becoming is generated by causet growth. In any case, however, the core idea of Broad's notion of Becoming—that the sum total of existence is always increasing—is embodied in causet dynamics.

There is another aspect of the causal set approach that is worth mentioning here. The proponents of this approach regard two finite-stage causet Becoming models $C_m = \{\emptyset \rightsquigarrow C_1 \rightsquigarrow \ldots \rightsquigarrow C_m\}$ and $C'_m = \{\emptyset \rightsquigarrow C'_1 \rightsquigarrow \ldots \rightsquigarrow C'_m\}$ as physically equivalent if $C_m =$

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C'_m. (For a concrete example, see Figure 3. Dots represent elements of causets, and a line between dots represents the fact that the corresponding elements are $\prec$-related with the direction of the relation indicated by height.) Their reason is rooted in the fact that the stochastic dynamics is introduced by means of the heuristic of an external time providing an ordered labeling of the birthing of events. But then they are at pains to

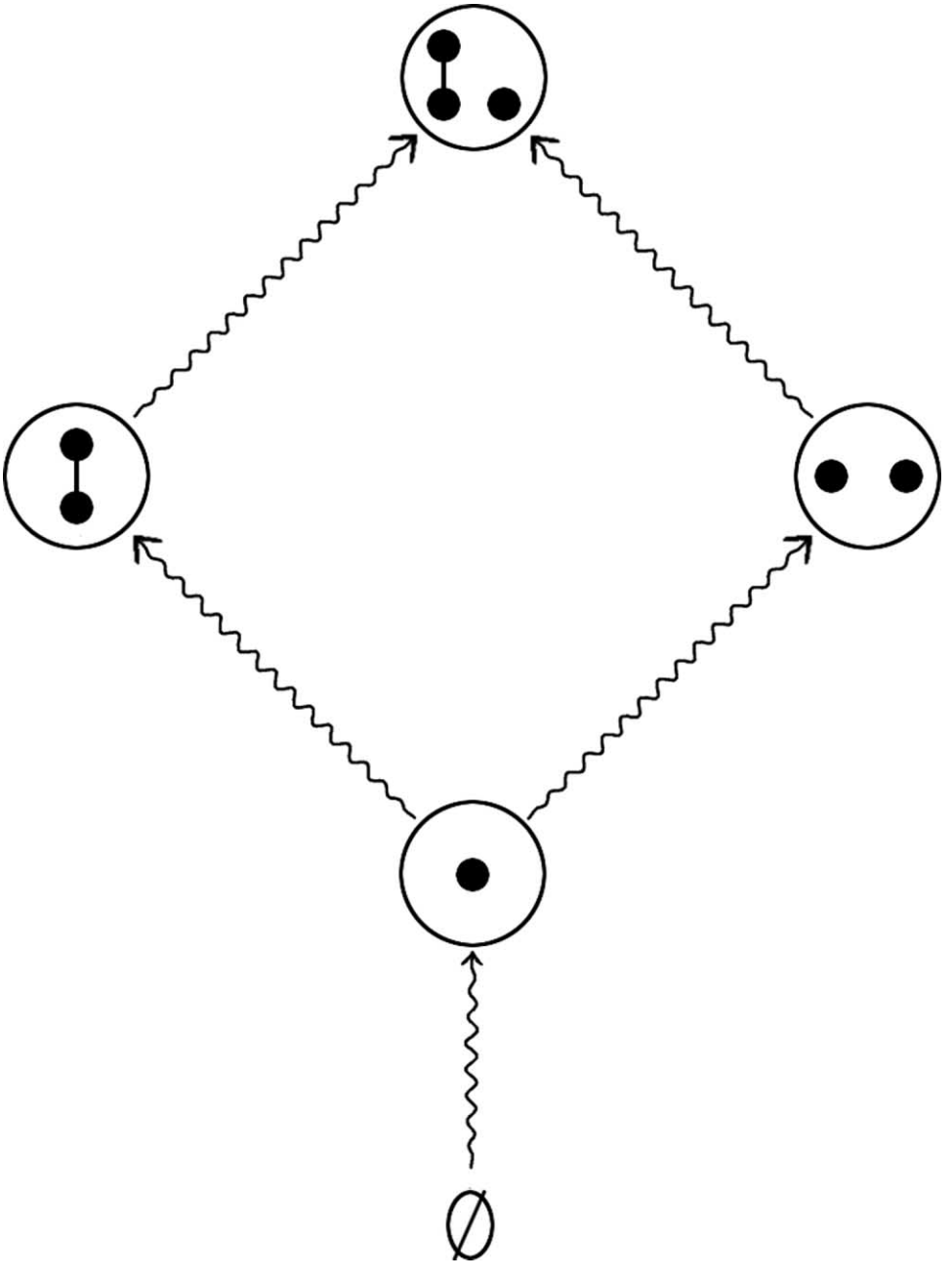


Figure 3 Stochastic Sequential Growth.

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emphasize that this external time does not carry any physical significance, and their way of underscoring the point is to assert the physical equivalence of the above C_m and C'_m and to require that the product of the transition probabilities for the two models is the same.[42] By extension, two infinite causet Becoming models C_∞ and C'_∞ should be counted as physically equivalent if for any m there is an $n \geq m$ such that $C_n = C'_n$. Causet Becoming would then be properly represented by equivalence classes of the original infinite Becoming models. This entails an indeterminacy of Becoming for non-unit equivalence classes, and the bigger the equivalence class the greater the indeterminacy.

But before becoming alarmed at the prospect of such an indeterminacy, it is worth asking whether the prospect results from an overcompensation for the heuristic of an external time. Since quantum aspects are ignored in classical sequential growth dynamics, it seems fair to ignore a many-worlds interpretation of the stochastic birthing process and to assume that at each stage only one of the possible alternatives is actualized. Thus, in Figure 3 the Becoming for the actual world is modeled by (say) the left-hand path—but not by the right-hand path—if the intermediate causet on the right is not in fact actualized. If it is legitimate to worry that the actualization/non-actualization distinction presupposes an external time, then the ability of the causets approach to deliver a model of Becoming is seriously undermined. But I see no reason for the causets proponents to give in to this worry—or more precisely, I see nothing in the causets approach that makes the worry any more legitimate than it is for Broad's model.

Needless to say, the picture of Becoming that emerges from the causets approach may change radically when the classical sequential growth dynamics is replaced by a quantum dynamics—at worst, Becoming could dissolve into a quantum foam, leaving no recognizable trace of Broad's original idea. Of course, it is open to proponents of Becoming to help themselves to the apparatus of stochastic birthing of events without linking the apparatus to quantum gravity or indeed to any programme in physics. But once such linkages are severed, the apparatus becomes a plaything of speculative metaphysics.

8. Conclusion

Even for self-confessed Blockheads such as myself, there is something undeniably attractive about Broad's idea of capturing the intuitively felt unfolding of time by means of growing block models. However, a surprisingly large number of problems crop up when one tries to construct block models, even in the friendliest environment of Newtonian spacetime, with absolute simultaneity; and the problems get progressively worse as one progresses from Newtonian, to special relativistic, to general relativistic spacetimes. Even when these problems admit remedies, there remains the lingering worry that the growing block models are parasitic on the block models, both in the sense that the construction of the growing block models illicitly relies on using chips off block models and in the sense that a condition of adequacy on growing block models is that they yield back block models as ideal completions of Becoming.

A contentious case arises when the growing block model does not fix a unique block model. Such cases occur, for example, for the worldline-relativized version of Becoming

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(Section 6) when the pieces of the growing block model correspond to past light-cone chips off general relativistic block models with observationally indistinguishable counterparts. For the Broadheads there is in such cases an ontic indeterminacy in the associated block models. And for them this indeterminacy is not only *not* disturbing but profoundly pleasing, since it underscores the ideality of block models. What the Broadheads find profoundly pleasing the Blockheads find profoundly unsatisfactory, since they can tolerate epistemic but not ontic indeterminacy in their models. How to settle this standoff is profoundly unclear.

It is appropriate here to comment on recent attempts to deflate the debate regarding *presentism* (only the present is real or exists), *past–presentism*[43] (only the past and present are real or exist), and *eternalism* (the future and past are as real as the present). For example, Dorato (2006) has argued that there is no genuine disagreement here and that the illusion of disagreement is generated by oscillating back and forth between tensed and tenseless senses of existence (see also Savitt 2006).[44] While I am not sure I agree entirely with this argument for deflation, I do agree that there no interesting ontological issue at stake in the way the debate is usually posed, since, as noted above, the participants typically agree about the spacetime structure (e.g. Minkowski spacetime or some other future-inextendible spacetime) and about the events that populate this structure. The only room for disagreement is over how to assign labels ('real', 'exists') to the events. There are many consistent conventions for such assignments; and even if one of the conventions turns out to be more natural or more preferable by some criterion, this makes no difference for the underlying ontology. Concomitantly, if a future-inextendible model is adopted as the model of our world, then the only sense of 'becoming' available is a thin and yawn-inducing one: by definition, events occur/happen, and the successive occurring/happening of events is becoming (see Dorato 2002, 2006).

By contrast, there is an ontological difference between a future-inextendible block model presupposed by the standard versions of eternalism and the growing block model as explicated here; in particular, a future-inextendible block model is not an element of a growing block model, and from the point of view of the latter the former 'exists' only as an ideal completion of the growing block. And this difference makes available a less yawn-inducing sense of becoming in growing block models—indeed, a sense that deserves to be capitalized to distinguish it from the thin sense. As noted in Section 4, there are ways to try to diminish the difference between block vs. growing block models as regards the reality of the future. But it is wholly implausible that the difference can be made to disappear; in particular, for any natural semantics 'The sufficiently distant future is literally unreal/literally does not exist' and 'The sum total of existence was less than it is now and will be greater' are true in every element of a growing block model but false in future-inextendible block models.

Apart from the hoary philosophical debates about the reality of the future, the growing block model is given new life by the causal set approach to quantum gravity. By providing a physical mechanism for producing growth and Becoming, this approach promises to transmute Becoming from a piece of speculative metaphysics to one of naturalized metaphysics. Furthermore, the causal set approach avoids any appearance

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of parasitism on block models, since it makes classical relativistic spacetime an emergent feature of Becoming, something Broad would have no doubt applauded. Unfortunately, the versions of causal set dynamics that have been studied thus far do not incorporate quantum interference effects and, consequently, the nature of Becoming and the emergence of spacetime in a full quantum version of the approach remain obscure.

Should the growing block model prove itself to be a coherent ontological alternative to the block model, one could still try Carnap-style deflation. The proposal would be that the (future-inextendible) block models and the growing block models should be regarded as different frameworks for describing the world, the choice between which is a purely pragmatic affair to be settled by such criteria as efficiency, fruitfulness, simplicity, etc. (see Carnap 1950). I do not disagree with such a proposal. But at the same time, I don't think that it ends the debate. The issue now becomes which framework lends itself best to the scientific enterprise of constructing a quantum theory of gravity, to the philosophical enterprise of understanding the notion of causation as production (i.e. causes bring their effects into existence), and the like. To call such issues pragmatic is not to dismiss their importance.

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Notes

[1] For a readable overview of 'static' vs. 'dynamical' conceptions of time and references to the literature, see Dainton (2001). The terminology is rather unfortunate. Those who defend the block model as an adequate representation of physical change, and as the basis for an explanation of our perceptions of change, will feel unfairly stigmatized by the 'static' label. However, some of the alternatives to the block model hardly seem to merit the label 'dynamical' (see Section 4). But these labels are so well entrenched it is pointless to try to displace them.

[2] As some commentators have noted, McTaggart would have done better to speak of *A-properties* rather than the *A-series* (what is the series and what is the ordering relation?). But the latter term has become standard in the literature, and I will continue to use it here.

[3] See the contributions to Oaklander and Smith (1994), and also Savitt (2001).

[4] The most sustained defence of the *B*-series account of change is to be found in Mellor (1998). For an attempt to account for the phenomenology of temporality within a block universe setting, see Butterfield (1984).

[5] For an account Broad's later views on Becoming, see Savitt (2002).

[6] It would be interesting to investigate the relation between the directionality of Becoming and the so-called arrows of time discussed under the heading of the problem of the direction of time, but that is a project for another occasion. There is an obvious linkage to the so-called electromagnetic arrow of time, since, presumably, the use of an advanced representation of the electromagnetic field is inappropriate if (as Broad maintains) future events are nonentities. For an overview of the issues involved with the electromagnetic arrow, see Earman (forthcoming).

[7] McTaggart died early in 1925.

[8] Indeed, I must confess: *Ich bin ein* Blockhead. But my purpose here is to give the Broadheads a run for their money.

[9] I will say that two spacetime models $\langle M, G_1, G_2, \ldots, P_1, P_2, \ldots\rangle$ and $\langle M', G'_1, G'_{2'}, \ldots, P'_1, P'_2, \ldots\rangle$ are isomorphic iff there is a diffeomorphism $d : M \to M'$ that is onto and $d^\star G_i = G'_i$ and $d^\star P_j = P'_j$ for all i and j, where $d^\star$ denotes the drag along by d. A diffeomorphism is a one–one map that preserves the differentiable structure. For sake of simplicity, assume that the manifolds are all C^∞ and, thus, that d is C^∞.

[10] The reader interested in the details can consult Earman (1989, ch. 2).

[11] Similarly, the relation $\prec$on $\mathfrak{N}$ defined by the condition that $\mathrm{n} \prec \mathrm{n}'$ iff n can be isomorphically embedded as a proper submodel of n′. That $\langle M, G_1, G_2, \ldots, P_1, P_2, \ldots\rangle$ is isomorphically embeddible into $\langle M', G'_1, G'_{2'}, \ldots, P'_1, P'_{2'}, \ldots\rangle$ means that there is a diffeomorphism d if M into M' such that $d^\star G_i = G'_i$ and $d^\star P_j = P'_j$ for all i and j.

[12] It would be suspicious to require that all of the elements of $\mathfrak{N}$ are isomorphic to a chip off one and the same Newtonian block. This suspicion was avoided above by requiring only that each element of $\mathfrak{N}$ is isomorphic to a chip off some Newtonian block or other. But the issue comes back later on in another form, namely, does the growing block model fix a unique block model? See below.

[13] I suspect that it can't be done without introducing a fifth dimension and thereby reawakening the embarrassments attendant to it.

[14] I find Broad's ordinary language analysis of this issue unpersuasive; see Broad (1959, 767–777).

[15] A Newtonian block model $\langle M, G_1, G_2, \ldots, P_1, P_2, \ldots\rangle$ is *stationary* if there is a one-parameter family of diffeomorphisms d_λ, $-\infty < \lambda < +\infty$, such that d_λ maps each plane of absolute simultaneity $T = \Delta$ to $T = \Delta + \lambda$ and $d^\star_\lambda\ G_i = G_i$ and $d^\star_\lambda\ P_j = P_j$.

[16] Relationists will charge that the alleged indeterminism in the above example is only an illusion fostered by a substantivalist conception of spacetime as a non-material medium in which physical objects are immersed. Formally the relationist claim would be that the block models $\mathcal{N}$ and $\mathcal{N}^*$ of Figure 1 are merely different representations of the same physical situation. The connection between spacetime substantivalism and Becoming will come up again in the following section.

[17] I.e. a means of identifying time slices across elements of $\mathfrak{N}$. Then $\precsim$ could be defined as follows: $\mathbf{n} \precsim \mathbf{n}'$ iff $\mathbf{n}$ can be isomorphically embedded as a submodel of n′ by a map that matches identical time slices. This relation is antisymmetric.

[18] Note that the ideal completion of a growing block models is not constructible by a Zorn's lemma argument.

[19] It is wholly implausible that by 'there is' Broad means 'there is at present', since this would turn his statement into the triviality that there is at present no event to which a present event is temporally precedent. Broad believed that one happy consequence of his doctrine of the literal unreality of the future was that it explained the 'impossibility of absolutely certain knowledge about the future' (Broad 1923, 73). His claim was that we can have certain knowledge of an event only if we are directly acquainted with it. But 'we can be directly acquainted only with *something*, not with a mere *non-entity*. On our view we cannot stand in the relation of direct acquaintance to future events, for the same reason which prevents us from robbing a Highlander of his breeks [*viz.* there are none]' (79). One might object that the same result can be obtained by accepting the reality of the future while requiring that direct acquaintance requires a causal connection and that there is no backward causation.

[20] That a relativistic spacetime M, g_{ab} is *time orientable* means that there exists an everywhere-defined continuous timelike vector field. Such a field induces a continuous division of the lobes of the null cones at every point of M into two classes. The choice of one of these classes as the 'future' lobes constitutes a time orientation. Block models face the problem of providing a basis for this choice, which is an aspect of 'the problem of the direction of time'.

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[21] These and other causality requirements for relativistic spacetimes are discussed in Hawking and Ellis (1973) and Wald (1994).

[22] For example, the Einstein static cosmological model is unstable; see Hawking and Ellis (1972, sec. 5.3).

[23] The most famous member of this class is the Wheeler–Feynman time symmetric, action-at-a-distance version of classical relativistic electrodynamics; see Wheeler and Feynman (1945, 1949). There are good reasons for discounting such theories on grounds of empirical adequacy. But they do serve to make the point that there is no *a priori* guarantee that future-truncated models are compatible with relativistic theories.

[24] More precisely, M, g_{ab} satisfies *strong causality* iff for any point $p \in M$ and any open neighborhood N of p, there exists a neighborhood N' of p contained in N such that any causal curve that leaves N' never reenters it. If it is required only that for any point $p \in M$ there exist an open neighborhood N of p such that any causal curve that leaves N never reenters it, then one gets the weaker requirement of future distinguishing (see below).

[25] See Arntzenius and Maudlin (2005) and Earman et al. (forthcoming).

[26] The underdetermination issue will rear its head in the next section.

[27] The necessary and sufficient condition for V^a to be hypersurface orthogonal is that it be irrotational, i.e. $V_{[a;b]} = 0$.

[28] See note 27.

[29] A relativistic spacetime M, g_{ab} is *globally hyperbolic* iff it is strongly causal and, for all $p, q \in M$, $J^-(p) \cap J^+(q)$, compact. A spacetime is globally hyperbolic in this sense iff it admits Cauchy surfaces.

[30] Here are some of the points on which I disagree with Tooley. 1) Analyzing our commonsensical judgments (or even worse, judgments tutored by analytical metaphysicians) is not a fruitful way to get at the deep structure of the world. 2) Causation is a folk notion, and trying to use it in interpreting theories of physics produces more mischief than enlightenment. 3) There aren't two versions of STR, one which posits that the one-way speed of light is the same in all directions and in all frames and one which doesn't. In fact there isn't any substantive issue here, since the effects in question are coordinate-dependent and since STR can be stated in a completely coordinate-independent fashion. 4) Understanding the Einstein-Rosen-Podolsky-Bell correlations does not require an Absolute Frame or Absolute Simultaneity.

[31] For example, Wald (1994, 360) writes, '[I]f $\phi : M \to M$ is a diffeomorphism, then (M, g_{ab}) and $(M; \phi^* g_{ab})$ represent the same physical spacetime'.

[32] For an overview of the issues involved, see Stachel (1993).

[33] 'A relative lapse of time … if any meaning at all can be given to this phrase, would certainly be something entirely different from the lapse of time in he ordinary sense, which means change in the existing. The concept of existence, however, cannot be relativized without destroying its meaning completely … A lapse of time … which is not a lapse in some definite way seems to me as absurd as a colored object which has no definite colors' (1949, 558n5).

[34] But at least the stipulation works. For the worldline-relativized version of Becoming discussed in the next section, the stipulation may not work.

[35] Note that the sense of immortality involved here does not entail that the observer lives for an infinite amount of proper time but only that she goes on living as long as the spacetime structure permits. An endpoint p of a worldline γ need not be a point on γ; rather the condition for being an endpoint is that any open neighborhood of the point intersects γ.

[36] For a relativistic block spacetime M, g_{ab} the event horizon of a timelike curve γ can be defined as the boundary of $J^-(\gamma)$ in M. In the example illustrated in Figure 2, the event horizon of $\hat{\gamma}$ consists of the null line L.

[37] See Hawking and Ellis (1973, sec. 5.2) for a description of this spacetime.

[38] For an overview of the evidence for and interpretations of accelerating expansion, see Carroll (2004).

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[39] This is a slight modification of Malament's (1977) definition, which uses the chronological past instead of causal past.

[40] In addition to the Markov condition, three other conditions, called internal temporality, discrete general covariance, and Bell causality are imposed. The first two of this trio will come into play below.

[41] If C is not regarded as a subset of C' (as would be the case if C_∞ is composed of isomorphism classes of finite causets), then $C \rightsquigarrow C'$ can be defined to hold iff there is an isomorphic imbedding $\iota : C \to C'$ such that $C' = \iota(C) \cup \{d\}$. This introduces an element of 'gauge freedom' into the description of the transition from C to C', since two such embeddings are regarded as physically equivalent if they are related by an automorphism of the child C'. Naturally, the transition probabilities for transitions associated with equivalent embeddings are required to be the same.

[42] See the discussions of 'discrete general covariance' in Rideout and Sorkin (1999) and Varadarajan and Rideout (2006).

[43] Sometimes called possibilism in the literature.

[44] For attempts to resist deflation, see Sider (2001, 2006). I think that resistance is futile if the block model is accepted.

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